

GRADE 12 PHYSICAL SCIENCES NSC PAPER 2 (CHEMISTRY - ORGANIC & EQUILIBRIUM)

Subject: Physical Sciences | Grade: 12 | Academic Year: 2024 | Assessment: Term 4 Examination

Document Type: Standard Examination Question Paper • Time: 2 - 3 Hours • Total Marks: 100 - 150

INSTRUCTIONS AND INFORMATION

- This document consists of all standard curriculum questions and study guidelines required for DBE CAPS moderation.
- Read through all questions and reference notes carefully before attempting solutions.
- Answer ALL questions according to the specific rubrics and curriculum guidelines provided.
- Number the answers correctly according to the numbering system used in this question paper.
- Write neatly and legibly. Work must show all calculation steps, units, and clear final statements.

SECTION A: CORE SYLLABUS CONCEPTS & QUESTIONS (Physical Sciences Grade 12)

QUESTION 1: NEWTON'S LAWS & KINEMATICS [25 MARKS]

1.1 A crate of mass 15 kg is pulled across a rough horizontal surface by a constant force of 60 N at an angle of 30° to the horizontal.

- 1.1.1 State Newton's Second Law of Motion in words. (2 marks)
- 1.1.2 Draw a labelled free-body diagram of all forces acting on the crate. (4 marks)
- 1.1.3 Calculate the magnitude of the normal force exerted by the surface. (3 marks)
- 1.1.4 If the coefficient of kinetic friction is 0.25, determine acceleration. (5 marks)

AUTHENTICATED CAPS EXAMINATION MATERIAL

Produced by Fusion High School Management System • Verified by Curriculum Head: Dr. T. Makola
This document is official study material aligned with the DBE National Curriculum Statement (South Africa).